# RATIONAL EQUIVALENCE IS PRESERVED BY EXTERIOR PRODUCT

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file intersection.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

*Editorial note.* Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 0 hits for this chapter (is-occupied.json, 2026-10-14); consistent with the duty manager's six-check dispatch.

*Source reference.* Original reference 038F: sources/stacks-chow/tags.txt, source lines 4234–4234. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original lemma 0AZM: sources/stacks-chow/tags.txt, source lines 13830–13830. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 00IX: sources/stacks-chow/tags.txt, source lines 695–695. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

*Source reference.* Original reference 02M1: sources/stacks-chow/tags.txt, source lines 3428–3428. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

**Lemma 0.1.** *Let $X$ and $Y$ be varieties. Let $\alpha \in Z_r(X)$ and $\beta \in Z_s(Y)$. If $\alpha \sim_{rat} 0$ or $\beta \sim_{rat} 0$, then $\alpha \times \beta \sim_{rat} 0$.*

*Proof.* By linearity and symmetry in $X$ and $Y$, it suffices to prove this when $\alpha = [V]$ for some subvariety $V \subset X$ of dimension $r$ and $\beta = [W_a]_s - [W_b]_s$ for some closed subvariety $W \subset Y \times \mathbf{P}^1$ of dimension $s+1$ which intersects $Y \times a$ and $Y \times b$ properly. In this case the lemma follows if we can prove

$$[(V \times W)_a]_{r+s} = [V] \times [W_a]_s$$

and similarly with $a$ replaced by $b$. Namely, then we see that $\alpha \times \beta = [(V \times W)_a]_{r+s} - [(V \times W)_b]_{r+s}$ as desired. To see the displayed equality we note the equality

$$V \times W_a = (V \times W)_a$$

of schemes. The projection $V \times W_a \to W_a$ induces a bijection of irreducible components (see for example Varieties, Lemma 038F). Let $W' \subset W_a$ be an irreducible component with generic point $\zeta$. Then $V \times W'$ is the corresponding irreducible

*Date*: stacks-project:intersection.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

component of $V \times W_a$ (see Lemma 0AZM). Let $\xi$ be the generic point of $V \times W'$. We have to show that

$$\text{length}_{\mathcal{O}_{Y,\zeta}}(\mathcal{O}_{W_a,\zeta}) = \text{length}_{\mathcal{O}_{X\times Y,\xi}}(\mathcal{O}_{V\times W_a,\xi})$$

In this formula we may replace $\mathcal{O}_{Y,\zeta}$ by $\mathcal{O}_{W_a,\zeta}$ and we may replace $\mathcal{O}_{X\times Y,\zeta}$ by $\mathcal{O}_{V\times W_a,\zeta}$ (see Algebra, Lemma 00IX). As $\mathcal{O}_{W_a,\zeta} \to \mathcal{O}_{V\times W_a,\xi}$ is flat, by Algebra, Lemma 02M1 it suffices to show that

$$\text{length}_{\mathcal{O}_{V\times W_a,\xi}}(\mathcal{O}_{V\times W_a,\xi}/\mathfrak{m}_\zeta \mathcal{O}_{V\times W_a,\xi}) = 1$$

This is true because the quotient on the right is the local ring $\mathcal{O}_{V\times W',\xi}$ of a variety at a generic point hence equal to $\kappa(\xi)$. $\square$